PH.45320 **FALL 2026**

Static Cosmology

Lecture 10: Equilibrium of Static Populations

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Overview

A stationary kinetic equation balances formation, transport, collisions, and removal of stellar populations.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$\partial_t f + v \cdot \nabla_x f - \nabla\Phi \cdot \nabla_v f = C[f] + S[f] - D[f] = 0$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$N(\tau) = \int R(t-\tau) P_{surv}(\tau) dt$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$\delta \dot{x} = \mathcal{E} e \delta f$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 10
- Population Equilibrium Tables
- Stellar Age Registry